

The formation of a RIVER CANYON

The Grand Canyon is a spectacular example of a gorge—a steep-sided mountain valley eroded by a river. Weathering processes have both shaped the inside of the canyon—producing features such as mesas, buttes, and spires that are often seen in arid and semiarid places—and helped make it wider (see panel, below).

A river requires a lot of erosive power to carve such a deep gorge. The Colorado owes its power to its steep gradient, which increases the speed at which the water moves along its channel, and to a relatively high sediment load, which enables the water to scour and deepen the channel.

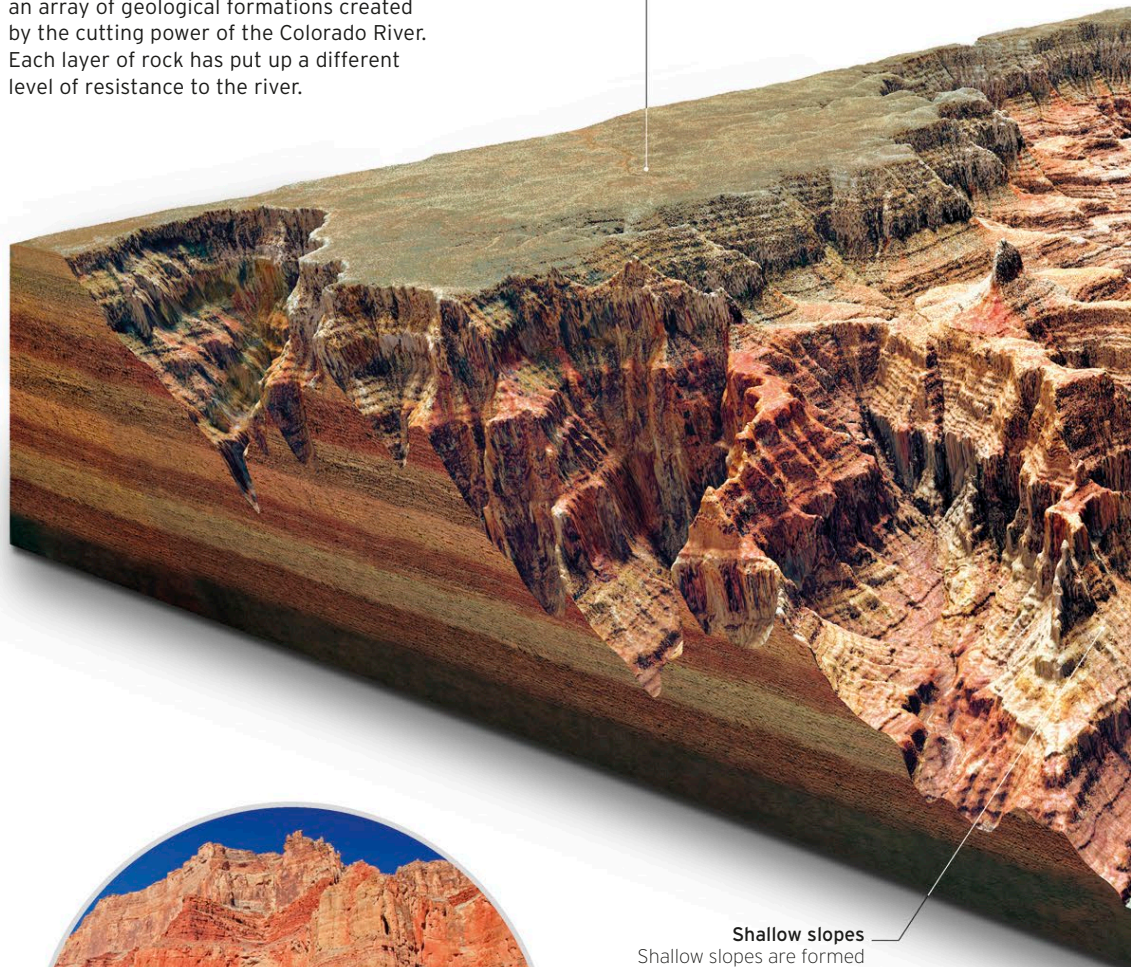
The Grand Canyon is a relatively young canyon carved into old rocks. It seems likely that the Colorado gained the power to carve the canyon when the Colorado Plateau was pushed upward by tectonic forces, steepening the slope of its bed. This sudden increase in a river's power is called rejuvenation.

► THE GRAND CANYON

This cross section of the Grand Canyon, at a point near the Powell Plateau, displays an array of geological formations created by the cutting power of the Colorado River. Each layer of rock has put up a different level of resistance to the river.

The South Rim

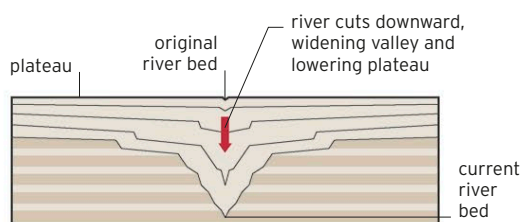
The South Rim tilts away from the canyon, so water falling on its surface does not flow over the edge into the Colorado.



Shallow slopes
Shallow slopes are formed when a river encounters a layer of soft rock, such as shale.

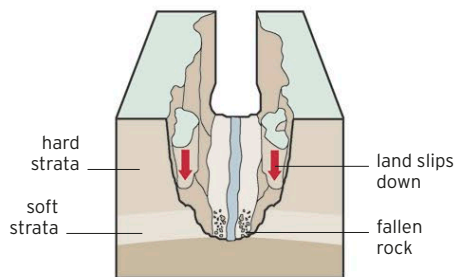
HOW THE GRAND CANYON WAS FORMED

The Colorado Plateau is made up of layers of sedimentary rock (or strata), including limestone, sandstone, and shale. The shape of the canyon is partly determined by how resistant these rocks are to processes of weathering and erosion.



DOWNCUTTING

The Colorado River has cut down progressively into the plateau. The canyon widens significantly whenever the river encounters a layer of relatively soft rock.



CLIFF COLLAPSE

The river is able to cut relatively easily through soft strata, undercutting cliffs above made of harder rock and causing them to collapse.



◁ COLORADO CONFLUENCE

The Little Colorado River is one of the Colorado River's largest tributaries. Tributaries add both water and sediment to the main channel, often altering the color of the water. The point where two rivers meet is called the confluence.

The **oldest rocks** exposed in the walls at the **bottom of the canyon** are almost **2 billion years old**